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ABSTRACT

This dissertation focuses on the design, analysis, and optimisation of microstrip antennas using a combination of electromagnetic simulations and machine learning (ML) techniques. Three types of antennas were created in COMSOL Multiphysics: a circular microstrip patch antenna with a coaxial feed, a planar inverted-F antenna (PIFA), and an 18 GHz rectangular microstrip linear array. These antennas were modelled, meshed, and simulated to extract key performance metrics such as S_{11} , VSWR, input impedance, gain, efficiency, and radiation patterns. The 18 GHz linear array served as the main case study for generating a large dataset through parametric sweeps, since full-wave simulations at high frequencies are computationally expensive and sensitive to meshing quality.

The simulation data was preprocessed, normalised, and used to train three supervised ML models: an artificial neural network (ANN), a support vector regression (SVR) model, and a random forest (RF) regressor. The ANN demonstrated the highest accuracy, achieving test R^2 values above 0.97 for most performance metrics and showing a strong ability to learn nonlinear relationships between antenna geometry and electromagnetic behaviour. The RF model also performed well, particularly for predicting S_{11} , VSWR, and impedance. The SVR model produced acceptable predictions but showed larger errors for directivity and gain.

To verify ML performance, the predicted antenna results were compared with COMSOL outputs for both known and custom geometries. The RF model, for example, achieved an average error of about 9.3% for a validation design, demonstrating that ML models can replicate full-wave results with adequate accuracy. Once trained, these models provide nearly instantaneous predictions, significantly reducing the time required for design exploration and optimisation compared with full COMSOL simulations.

Overall, this work shows that ML-based surrogate modelling is a practical and efficient approach to antenna design, especially when handling large parameter spaces and high-frequency structures.

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1. INTRODUCTION:

1.1 Background:

In the world of wireless communication, antennas play a crucial role as technologies continue to grow and evolve, which in turn demands higher performance. This dissertation focuses on a microstrip antenna, which serves as the basis for AI/ML-based optimisation. Microstrip antennas are widely employed in a broad range of applications, including 5G and 6G communication systems, IoT devices, satellite links, radar platforms, aircraft communication, and millimetre-wave systems. Despite advantages, single patches suffer from inherent limitations: low gain, narrow bandwidth, limited radiation efficiency, and strong dependence on substrate and feed design. Microstrip antennas offer several advantages and can achieve high gain; however, when extended to array configurations, additional feed networks are required, which can complicate the design process. [1]

An antenna array is a collection of multiple antennas that are operated in proximity to one another and are designed to be coordinated as a single element. It leverages the interference pattern of the fields radiating from the various elements to achieve a desired radiation pattern, and it is often enabled with tighter directivity than could be achieved with a single-element design [2]. It is also called a group of radiators or an array of antenna elements. By controlling the phase and magnitude of the input signal assigned to each antenna element, as well as the number and arrangement of elements in the array, it is possible to steer the radiation pattern in a desired direction with a preferred level of gain. As the number of elements in the antenna array increases, the feed gets more complicated. Thus, it is desirable to isolate the feed network structure from the antenna structure. [3]

Microstrip antenna arrays (MSAAs) can be implemented using either corporate feed networks or simpler linear array configurations. Linear arrays are advantageous because they reduce feed complexity and minimize insertion losses, and their performance can be controlled through frequency-based beam steering. However, microstrip antennas inherently suffer from limitations such as narrow bandwidth and low gain, which become more pronounced in high-frequency communication systems. [4] They provide high gain and are planar in nature as well, but feed networks are needed for the antenna arrays, which can complicate the antenna design. Microstrip antenna arrays (MSAA) can be formed using a corporate or linear array configuration. Linear arrays have the advantage of a plain design, which is not as complex as corporate arrays, and linear arrays have reduced feed complexity losses and beam steerability with variation of the frequency; however, microstrips have the disadvantage of low bandwidth and low gain [5]

Antennas have a lot of parameters, such as length, width, feed position, substrate, feed length, etc., which can be calculated using the design-based formulas, but the antenna should be modelled using the software platform that employs the Finite Integration Technique (FIT) to solve Maxwell's

equations because those equations can't be solved using pen and paper, which could affect behaviour, including wave propagation, reflection, and diffraction within the antenna structure. [6] In this work, COMSOL Multiphysics is used; it is a physics and engineering simulation package based on the finite element method (FEM), which allows detailed analysis of wave propagation, reflection, and diffraction within the antenna structure. The FEM-based solver in COMSOL enables full-wave analysis of the antenna's performance, including near-field and far-field behaviour, over a wide range of frequencies [7]

High-frequency antenna design like MM-wave and 5G/6G range becomes more difficult as the wavelength becomes smaller, which means small changes in the geometry, which in turn force the COMSOL to mesh the elements finer, as it may detect low-quality elements which cause wrong and sometimes even wrong output and While developing antenna models, some issues were encountered in the process of generating the mesh in the COMSOL environment. Geometric variations led either to the deterioration of the meshes or failed attempts at meshing, especially in higher frequency simulations where smaller element sizes are needed. It entailed remeshing the full geometry, which added complexity and intensity to the computational tasks. When it came to solving a problem requiring meshes with smaller elements to replicate the electromagnetic waves properly, solving the problem took from hours to a few days.

This is where optimisation using AI/ML models comes in handy, as they help to predict possible and near-optimal outcomes once they are trained using simulation and parametric sweep data. Many research works have used models like artificial neural networks, random forests, etc., to find suitable geometric values for antenna designs, reducing the number of full-wave simulations needed and helping to speed up the overall design process.

1.2 Problem statement:

In antenna design, the performance of an array depends on many interacting parameters, such as patch, substrate, feed position, feed length, and element spacing, and these are nonlinear. A full wave could be time-consuming, especially if you need a larger dataset and a fully-fledged working antenna without error. In practice, a single design study can take from several hours to multiple days to complete the There is a problem, which is a lack of proper datasets on antennas or antenna arrays, so any datasets required for the ML models must be generated manually from simulations. As a result, there is a clear need for an approach that can use COMSOL-generated data to build fast surrogate models, such as AI/ML models, that can predict antenna performance more quickly and support the optimization of the design space.

1.3 Aim and objectives:

This dissertation aims to optimize single-patch areas generated using COMSOL Multiphysics simulations. The main aims of this work are:

To design and simulate several microstrip antennas in COMSOL Multiphysics.

- The process involves running parametric sweeps in COMSOL over various combinations of design parameters to generate a large dataset, which is necessary for ML models to learn effectively.

- Then preprocess the simulated data by handling any missing values and removing duplicated values.
- After preprocessing the dataset, split the final dataset into three parts each for training, validation, and testing sets according to the size of the available data, for example, 70% for training, 15% for validation, and 15% for testing, as used in this work.
- Choose suitable ML models based on the type of data and then tune these models using optimisation algorithms to achieve good accuracy.
- After training and testing, use the ML models to predict the desired antenna performance outputs and then redesign the antenna in COMSOL to check whether the predictions from the ML models are correct. [8]

The remainder of this dissertation is organised as follows. Chapter 2 reviews the relevant literature on microstrip antennas, antenna arrays, and the use of machine learning in antenna design. Chapter 3 describes the modelling, data generation, and development of the ML models in COMSOL Multiphysics. Chapter 4 presents the simulation and ML results, while Chapter 5 discusses these results. Finally, Chapter 6 summarises the work and discusses the possible future developments.

2. LITERATURE REVIEW

2.1 Microstrip antennas and related RF structures:

A microstrip antenna is also called a patch antenna, and this antenna produces a directional pattern. To put it in a better way, it can share a signal in a particular direction. This antenna consists of a patch, the ground dielectric constant of the substrate, and the thickness of the substrate. Microstrip antennas are typically small and lightweight, and they can operate at multiple frequencies. They support double or multiple frequency operation; however, a single patch produces low efficiency due to its conductor and dielectric losses, and it has a narrow bandwidth and low gain, and it can produce a better performance when used for short-range or high data rate applications [9]. Due to these disadvantages, several patches are combined to form an antenna array to improve the gain and radiation pattern. The following concept is discussed further below in the sections

2.1.2 Antenna arrays:

An antenna array is made up of a group of antennas working together as one under a single signal. This is used in cases where the radiation pattern of a single antenna is suboptimal in performance and gain or in cases where suppression of a specific direction's interference is required [11]. As a result, the use of antenna arrays is beneficial in 5G/6G communication, radar technology, satellites, and other areas with high requirements for increased gain and directivity. The use of an antenna array requires a feed network for energy distribution among the elements. For a liner or series arrangement, the array is connected in line with a single transmit line, with each component picking up energy from the incoming wave and transferring the remaining energy downstream to other components. This design is simple, space-efficient, and economical, but the design is bandwidth-poor with a frequency sensitivity problem. A corporate feed, on the other hand, transmits power by means of components such as T-junctions, enabling each antenna element to be separately powered. This form of feeding is associated with a wider bandwidth and

better impedance matching but is related to a larger area due to the increased feed lines. This is because a larger number of splits and bends can be sources of loss within the feed network itself.

2.1.3 Circular Microstrip Patch Antenna with Coaxial Feed:

COMSOL Multiphysics was used to simulate the performance of a circular microstrip patch antenna that resonates at 5.8 GHz based on an online design, as illustrated in **Figure 1**. The design consists of a circular copper patch of radius 0.773 cm placed on a dielectric substrate with $\epsilon_r = 3.55$ and thickness close to 0.8 mm and has a complete ground plane at the back. The antenna is fed by means of a coaxial probe where the inner conductor protrudes through the ground and the substrate and attaches to the centre of the patch 0.216 cm from the centre to ensure that it has an input impedance close to 50Ω , with the other end being connected to the ground plane. On the other hand, the outer conductor connects the ground plane. The patch, ground, and dielectric substrate are surrounded by an air domain enclosed by absorbing boundaries (PML) to simulate free space radiation. A simulation was run at 5.8 GHz to determine the S-parameters, gain, VSWR, and radiation patterns of the circular microstrip patch antenna. [13]

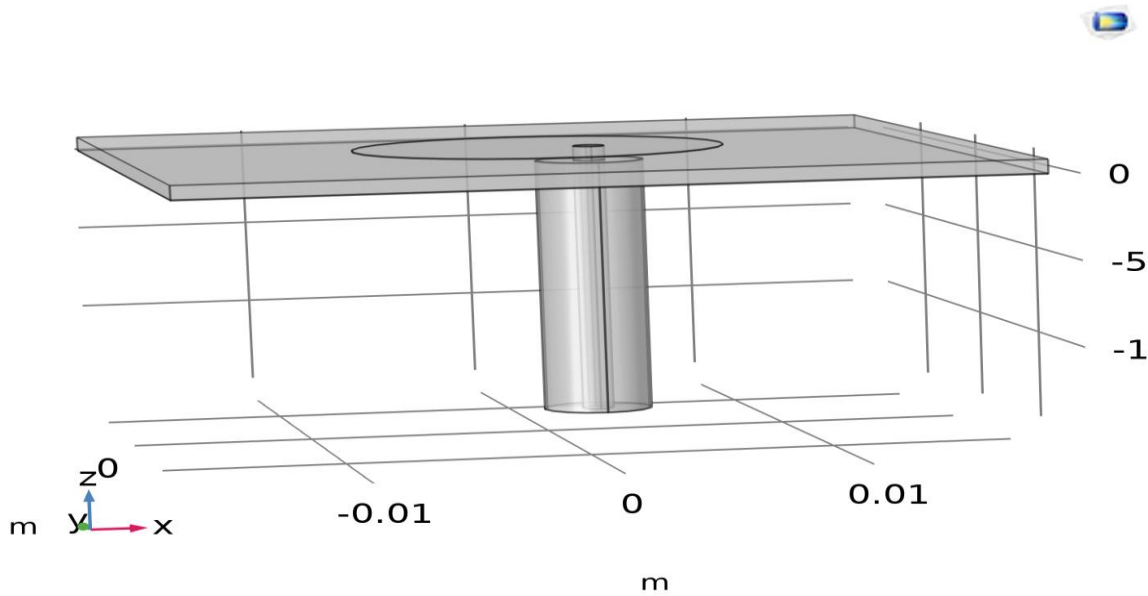


Figure 2.1: Circular Microstrip Patch Antenna with Coaxial Feed

2.1.4 Dual-Port Planar Inverted-F Antenna (PIFA):

In modern wireless communication technology, hand-portable devices such as mobile phones and tablets need antennas able to support high data rates with small dimensions. In these regards, multiple-input multiple-output (MIMO) technology solves the above challenge by using several antennas to increase data rates and reliability without needing any additional bandwidth. To overcome multipath fading, antenna diversity technology can be applied, using space diversity, pattern diversity, or polarisation diversity, or any combination thereof. In PIFA technology, there is the planar inverted F antenna (PIFA), generally applied in portable devices because of low profiles and simplicity of mounting inside the devices' housings. In earlier designs of PIFA technology applied in MIMO or diversity antennas, discrete PIFAs spatially separated and having

heights less than optimal in very thin devices generally demonstrated impedance bandwidths around 200 MHz. In reference [14], there was the proposal of a dual-port PIFA, consisting of a single resonating plate having two feed points positioned on both sides of the antenna, orientated orthogonally to each other. In these designs, there was both polarisation diversity and pattern diversity, requiring much wider bandwidth than other antennas at approximately 800 MHz wide, ranging from 2.1 to 2.9 GHz, uniformly covering both 2.45 GHz Bluetooth/WLAN and LTE/WiMAX communications [14].

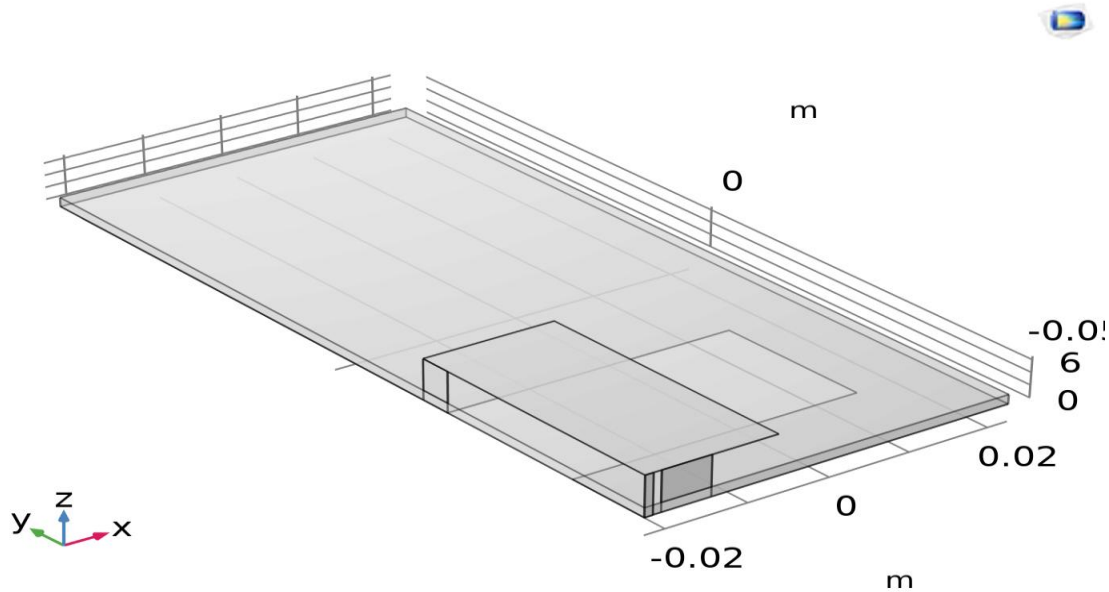


Figure 2.2. Dual-Port Planar Inverted-F Antenna (PIFA)

To begin, a compact dual-port planar inverted F antenna (PIFA) was created and modelled using COMSOL Multiphysics, having incorporated aspects from the example in [14] and further supplemented by real-world information from the tutorial video in [15], as shown in Figure 2. In the structure, there is a rectangular top layer placed above a ground plane, both separated by a dielectric layer with $\epsilon_r = 3.55$, $\tan \delta = 0.0021$, and about 0.8 mm thick. Both the radiator layer and the ground planes are 35 μm thick. There are two feed layers placed at both sides of the upper layer, and both are connected to the ground planes to form two ports perpendicular to each other to provide polarisation and pattern diversity. It was modelled in the range 4.5–7.1 GHz to analyse its resonant characteristics. The ambient medium was modelled from the analysis in COMSOL.

2.1.5 Rectangular Linear Microstrip Patch Antenna Array

In high-speed wireless communication, larger frequency-band millimetre waves have attracted much attention due to their capability of achieving ultra-high data rates with low latency. The 18 GHz band offers an attractive solution for the future of 5G and beyond; because of its small wavelength, large bandwidth can be obtained, and small antennas can be developed. To meet the growing requirements of ultra-high-definition TV and Internet of Things (IoT) communication, linear designs comprising antenna arrays are usually adopted to optimize the performance based

on beamforming and spatial diversity. In reference [16], an 18 GHz linear microstrip patch antenna array was presented and studied. It was shown that the designed antenna was able to produce a beam suitable for today's communication requirements using an RO3003 substrate [16]. Figure 3 shows the model of the linear microstrip patch antenna rectangular array considered in the analysis. It consists of six identical rectangular patch elements placed in line on a thin substrate of RO3003 with a complete ground plane at the bottom.

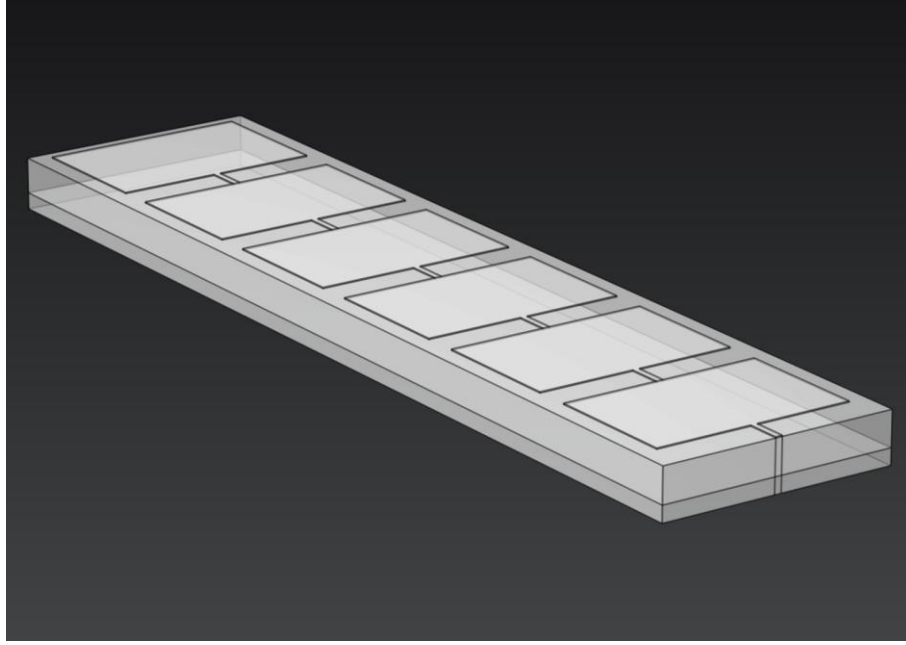


Figure 2.3: Rectangular linear microstrip patch antenna array at 18 GHz

In the current research, the same rectangular linear microstrip patch array of 18 GHz was simulated in COMSOL Multiphysics, as depicted in Figure 3, from reference [15]. The size of the antenna was found to be approximately 29.5 mm x 7 mm, and the gaps between the patches are filled with thin microstrip transmission lines that excite all six elements using the 50 Ω edge feed. The design creates constructive interference in the main beam and increases the gain of the antenna compared to the individual element. The model in COMSOL Multiphysics uses the same parameters as in reference [16] to ensure that all simulated antenna parameters utilise the same parameters as in reference [16] to ensure that all simulated parameters of the antenna match the already published data. Machine-learning algorithms can report and analyse S-parameters, VSWR, gain, current distribution, and far-field radiation patterns from the COMSOL Multiphysics model in reference [15]. Patterns Reference [15] can be reported and analysed from the COMSOL Multiphysics model in reference [15] and can be reported and analysed using machine-learning algorithms.

2.1.5 Summary of Antenna Structures Used in This Study

In this work, COMSOL Multiphysics is used for the simulation of three different antenna designs. To start, a circular microstrip patch antenna with a coaxial feed was modelled at 5.8 GHz. The layout consists of a circular copper patch on a dielectric substrate with a ground plane and a coaxial probe feed, and it basically helps in the analysis of a basic single-element microstrip radiator and the extraction of basic EM parameters, such as S-parameters, gain, and radiation patterns.

Second, a compact dual-port planar inverted-F antenna (PIFA) was developed based on an existing dual-port design. The dual-port PIFA has a single top layer with two feed locations positioned diagonally opposite corners to support polarisation and pattern diversity in the 2.1–2.9 GHz band. The PIFA model represents how a dual-port/diversity antenna works, although it was not included in the dataset of parametric sweeps in the machine-learning model. These three antennas offer single-patch, array, and dual-port scenarios to conduct EM analysis, and the remaining 18 GHz microstrip array represents the chief source of data collection for the machine-learning models developed in this work.

Thirdly, a linear rectangular microstrip patch antenna array working at 18 GHz was recreated based upon the design published in [15]. In the cited design, the antenna array consists of a linear series of six rectangular microstrip patch elements fabricated on an RO3003 substrate using a microstrip line feed, resulting in a compact directive antenna that can function effectively in the 5G/Ku band. For the new antenna design, we analysed different shapes and sizes, and from the results, we created the main data set for the machine-learning models in this dissertation.

2.2 Machine Learning in Antenna Design

ML models almost work in the same pattern. First, a set of antenna designs is created and simulated using an EM solver, and then, with different combinations based on the user needs, a parametric sweep is computed for a dataset. After computation, the dataset is extracted, cleaned, and divided into three parts: training, validation, and test sets. After training, the best model can be used to predict performance for new parameter combinations or to support optimisation by quickly evaluating many candidates and only simulating the most promising ones in COMSOL. In this dissertation, ML is used to predict key performance metrics of the 18 GHz microstrip array based on its geometric and material parameters. Three supervised regression models are considered: an ANN, an SVR model, and a random forest regressor. The end goal is to predict the best geometry using the outputs like S11 parameters, gain, VSWR, etc.

2.2.1 Machine Learning Models for Antenna Prediction

Artificial Neural Network (ANN): An Artificial Neural Network (ANN) represents a mathematical model that imitates the human brain's cortical structure, in which data processing takes place and signals are delivered to biological cells. In the analogous model, an ANN uses artificial cells rather than biological cells to simulate highly nonlinear relationships, such as those presented with near characteristics, to learn how output changes with geometries; the model in the dissertation utilises the ANN model [17].

Support Vector Regression (SVR): Support Vector Machines (SVM) are popularly used machine learning algorithms, and Support Vector Regression (SVR) can be considered as the regression counterpart of the SVM family. The basic idea of the SVR algorithm is to plot an appropriate function to predict the output with most of the data lying in or close to a fixed margin (epsilon). Data lying in the margin does not result in any penalty; however, data lying beyond the margin represents greater prediction deviations [8]. Kernel functions, like the radial basis function (RBF), handle the non-linear relationships between the inputs and outputs.

Random Forest (RF): It is based on three decision techniques and uses an ensemble learning

algorithm to make predictions by merging many decision trees in the model rather than using individual trees. Each tree in the model makes slightly different predictions, and usually, these predictions are obtained by averaging. It can predict how changes in the parameters of geometry and material affect other parameters, like S11, gain, or VSWR. It can help in understanding the level of importance of parameters in the prediction

3.

Methodology:

3.1 Antenna Modelling in COMSOL

In this chapter, the modelling and analysis steps used in this project are described. All antenna structures were implemented in COMSOL Multiphysics using the Electromagnetic Waves, Frequency Domain interface. Three antennas were modelled: a circular microstrip patch at 5.8 GHz, a linear microstrip patch array at 18 GHz and a dual-port planar inverted-F antenna (PIFA). The 18 GHz ring models, while the circular patch and PIFA are used to study additional antenna behaviours and support the results and discussion.

3.1.1 Circular Microstrip Patch Antenna

The first antenna model is the circular microstrip patch antenna, which will operate at 5.8 GHz. It comprises a circular copper patch of radius 7.73 mm located on a dielectric substrate of relative permittivity 3.55, thickness 0.81 mm (32 mil), and loss tangent 0.0021.

A complete copper ground plane is placed on the back of the substrate. Excitation of the antenna is through a coaxial probe feed where the inner conductor passes through the ground and substrate and is connected to the patch at a feed point offset of 2.16 mm from the centre. Coax feed has an inner diameter of 1.3 mm and an outer diameter of 4.35 mm while the feed length is 15 mm to assure a good impedance match for 50 Ω . The PML with a thickness of around 0.15λ is utilized to absorb outgoing waves and simulate free space radiation and terminates the surrounding air box. The simulation was carried out in the frequency range of 4.5 GHz to 7.1 GHz by using COMSOL Multiphysics' Electromagnetic Waves, Frequency Domain interface. S11, VSWR, gain, and radiation pattern were extracted from the results to study the antenna performance and validate its resonance at about 5.8 GHz.

3.1.2 Dual-Port Planar Inverted-F Antenna (PIFA)

The second model of the antennas considered in our work is the dual-port planar inverted F antenna. It consists of a rectangular top radiator placed above a ground plane using a dielectric substrate with a thickness of 6 mm and a permittivity of 2.2. The radiator size is 35 mm $\times$ 15 mm, while the ground plane size is 50 mm $\times$ 40 mm. It is compact and portable in nature. There are two shorting plates and two feed points, which are placed on opposite sides of the top plate, at the left edge and at the right edge, creating two ports that are placed at 90° separation to ensure polarization and pattern diversity. They are coaxial probes with feed heights same as the substrate thickness of 6 mm. They can excite ports separately to measure the mutual coupling between the ports. The band of operation of this PIFA is in the range of 2.1–2.9 GHz, providing an impedance bandwidth of about 800 MHz, suitable for use in Bluetooth, WLAN, and LTE/WiMAX. In the current analysis, the PIFA designs have been tested in terms of parameters like S11, S22, S21, VSWR, and far-field radiation pattern characteristics. The PIFA serves as a benchmark design in diversity and MIMO

antennas; in fact, it was not included in the machine-learning database since its task was merely analysis-orientated.

3.1.3 Rectangular Microstrip Linear Array (18 GHz)

In this paper, the first antenna model developed is a rectangular microstrip linear array resonating at 18 GHz in support of high-frequency requirements relevant to 5G and millimeter row on a substrate of RO3003 material, with $\epsilon_r = 3$, a height of 1.574 mm, and a copper ground plane. The size of the linear array is approximately 29.5 mm $\times$ 7 mm, with all antennas designed to resonate at 18 GHz.

The elements are coupled together using thin microstrip lines in a series-fed pattern, with the 50- Ω input feed line placed at one end of the array. In addition, the appropriate phase excitation of all six elements in the planar array has been ensured using the feed line. The surrounding medium has been assumed to be an air box with the Perfectly Matched Layer (PML) boundary condition.

Frequency domain analysis was performed using COMSOL Multiphysics from 16 to 20 GHz to test the antenna performance. The main parameters considered in the analysis included S11, VSWR, gain, and far-field pattern. The same model was also used to change key design factors like substrate height, patch size, and feed width, which helped gather data for training parameters, such as substrate height, patch size, and feed width, to produce data needed to train the other machine learning algorithms in the dissertation.

3.3 Meshing and Numerical Considerations

All antenna models in this project used a **user-controlled mesh** in COMSOL Multiphysics. Instead of the default physics-controlled mesh, the element size was set to “**Coarser**” to reduce the total number of elements and make the simulations more manageable. Mesh was used in these regions and produced a more regular layered mesh, which is suitable for flat microstrip structures. Even with a coarser global setting, the high operating frequency, especially for the **18 GHz array**, still required many elements to keep the solution stable.

In several cases, COMSOL displayed warnings such as “**low minimum element quality**”, mainly near tiny edges or narrow gaps in the geometry. Occasionally the meshing step failed completely when the edges were too small compared to the chosen element size. When it happened, the mesh settings had to be adjusted, for example, by slightly refining the swept mesh, changing local element size limits, or modifying the geometry so that the edges were less sharp and the sweep direction was simpler.

Using a coarser mesh helped to reduce the simulation time, but there was always a trade-off between speed and accuracy. These practical meshing problems and long run times were one of the reasons for introducing **machine learning** later in the project so that antenna performance could be predicted without having to repeat slow full-wave simulations for every new design.

3.4 Dataset Generation

For the development of the AI and ML models, a parametric sweep was performed using COMSOL Multiphysics, keeping the rectangular microstrip linear array as the baseline model. The aim of the parametric sweep was to produce data about the wide range of antenna performance characteristics by varying the key geometrical and material parameters. Parameters considered included patch length (L_p), patch width (W_p), substrate height (h_s), feed width (W_f), feed length (L_f), and element spacing (S), as shown in Table 3. Each of these parameters was varied from three to five discrete points, giving rise to several hundred distinct designs. For every design, a frequency sweep analysis was performed in COMSOL from 16 to 20 GHz, and data was collected regarding S_{11} (return loss), VSWR, input impedance, directivity, and gain.

All outputs from the simulation data were saved in the COMSOL Results -> Export -> Data format in CSV file format. These datasets were further processed in Python to eliminate any repeated or truncated data, and any missing numerical data was estimated using the average of the surrounding data points. Further processing involved normalising and splitting the data into three subsets: 70% of the data for training, 15% of the data for validation, and 15% of the data for testing.

In total, the parametric sweep resulted in about 2,700 data points, which can then be used to train and test the ML algorithms (ANN, SVR, and Random Forest). The data allowed the algorithms to learn the behaviour related to small parameter changes in the geometries or material characteristics to predict the optimal geometries without having to run further full-wave simulations.

3.5 MACHINE LEARNING MODEL DEVELOPMENT

3.5.1 Data Processing and Split

COMSOL's parametric sweep analysis of the linear microstrip array at 18 GHz provided the data for these machine learning algorithms. operation. The corresponding parameters defining the output are related to the performance characteristics of the antenna, such as S_{11} , VSWR, gain, directivity, and impedance. Before the model was trained, the raw data from COMSOL in the form of CSV files was further cleansed in Python. Duplicated data entries in the raw data were eliminated, and any missing data was either removed or made accurate through direct interpolation where necessary. The cleaned dataset comprises about 2,700 data entries.

the model, it does not need a scaler like the other two models. Later, the data was split evenly into three subsets for all models. These subsets included 70% data for the training of the model, 15% data for validating the model, and 15% data for testing the model. The data was split in such a manner because the test data helps in checking the effectiveness of the model in an unbiased manner.

3.5.2 Artificial Neural Network (ANN) Model

The first model based on machine learning used in the present work is Artificial Neural Networks (ANN) optimized for supervised regression analysis. The ANN developed in the task is designed to learn the non-linear association between the parameters of antenna designs and the corresponding performance factors obtained because of COMSOL analysis. The model takes 11 input variables including ten dimensional factors and frequency and gives out eight predicted values such as S11 at the design frequency, gain, VSWR, and other performance factors.

The ANN model consists of a fully connected feed forward network with three hidden layers. This model contains 20, 15, and 10 neurones in that, in order. tanh activation function was used on the hidden node to learn from the non-linear. The weights in the network are optimised using the LBFGS solver due to its suitability in medium-sized data sets..

The model is trained using normalised data; the role of the validation data in monitoring error during the training process can help in overcoming the issue of overfitting. Hyperparameters are tuned using the grid search algorithm in conjunction with 5-fold validation, where the parameters include the number of neurones in the layer, the activation function, and the regularization parameter. The model that performs the best gives an R-squared value close to 0.985 in the test data, suggesting that the ANN model can explain about 98.5% of the

variation in test data. The trained ANN model can predict the performance of the antenna with new parameters without requiring further COMSOL simulations.

3.5.3 Support Vector Regression (SVR) Model

The second model used in this research is Support Vector Regression (SVR). Support Vector Regression can be viewed as the counterpart of Support Vector Machine in the context of regression analysis, suitable to be applied when the dataset isn't huge but the correlation between inputs and outputs is nonlinear in nature. To ensure comparison can be made in an easier manner, the same input variables and corresponding output labels as in the ANN model are used.

The SVR model is written in Python using the "scikit-learn" library, with the kernel specified as the "radial-basis function" or "RBF." Standard Scaler scaling is applied to the output as well as the input. The main parameters in the SVR algorithm, namely "C," "epsilon," and "gamma," are specified using the "grid search" process with "5-fold" "cross-validation." An optimal setting in such cases can be specified as "C=10," "epsilon=0.1", and "gamma='scale'". e of the SVR model can be tested using the validation dataset and test dataset, using the same error criteria as in the ANN model. The final model, in the case of SVR, slightly performs less than the ANN model, but it correctly identifies the behaviour of the antenna based on the changes brought about by the parameters. The test R^2 value in such cases ranges from 0.93 to 0.95.

3.5.4 Random Forest (RF) Model

The third model tested in the study is the Random Forest regressor model. The model is an example of an ensemble learning model where the outputs of several decision trees are combined to improve

the reliability of the model. We use the same target outputs as the other two models, the ANN model and the support vector regressor model. Both the training process and the evaluation are based on the same data sets as in the other two cases, meaning the RF is trained and tested using the same data sets as the other two models. In addition to providing predictions, the Random Forest provides information about the feature importances in the form of so-called feature importance scores, which help reveal the most important parameters of the antennas affecting S_{11} , gain, and so on. Generally, the test R^2 values The performance metrics for the Random Forest (RF) model are around 0.97, indicating that its performance is intermediate compared to indicating that the model performance is in between the other two models.

3.6 Evaluation Strategy and Metrics

Each of the three algorithms—Artificial Neural Network (ANN), Support Vector Regression (SVR), and Random Forest—uses the same data split (70% for training, 15% for validation, and 15% for testing) when being trained and tested. The training data enables the algorithms to learn, the validation data helps in determining optimal parameters and identifying any signs of overfitting, and the test data is exclusively used to evaluate the results.

There are standard regression analysis metrics used to judge the models:

- R^2 (Coefficient of Determination): Indicates the proportion of the variance in the target data that the model accounts for.
- MSE (Mean Squared Error): It calculates the average squared difference between actual and predicted values.
- RMSE (Root Mean Squared Error): The square root of MSE, in the same units as the target variable.
- MAE (Mean Absolute Error): Average absolute difference between predictions and actual values.
- Percentage error: It represents the relative error in terms of the actual value.

These parameters are calculated based on the test data of corresponding models and presented in Chapter 4. By assessing ANN, SVR, and Random Forest algorithms using the same dataset and the same parameters, comparison between these algorithms allows defining which model can be most appropriately applied to predict the performance of the 18 GHz microstrip array antenna as well as to reduce the number of full-wave analysis tasks in future design processes.

4. Results & Discussion:

4.1 Circular Microstrip Patch Antenna

A. Reflection coefficient

The simulated reflection coefficient S_{11} plotted in Figure 4.1.1 for the 4–7 GHz band. The curve shows a clear resonance close to the design frequency, with a minimum value of about -22 dB

around 5.7–5.8 GHz. Within this region, S11 stays well below -10 dB, which indicates good impedance matching between the patch antenna and the $50\ \Omega$ feed line. Outside the resonant band, S11 rises towards 0 dB, showing that most of the power is then reflected and the antenna is no longer efficiently excited.

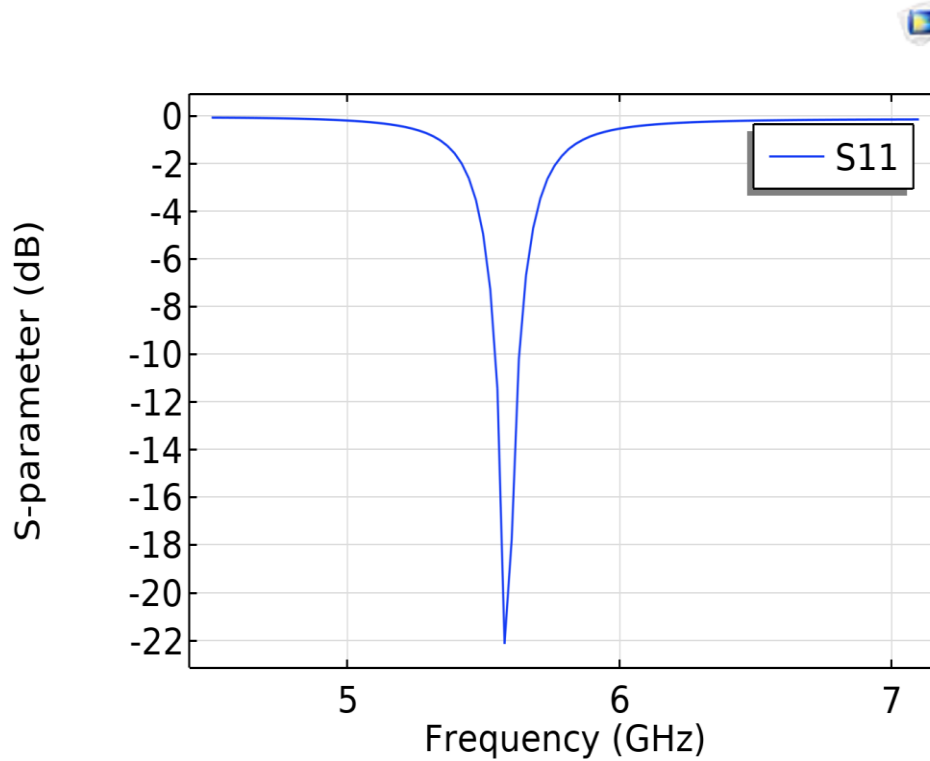


Figure 4.1.1

B. Impedance matching (VSWR)

The corresponding VSWR for Port 1 is shown in Figure 4.1.2. At the lower and upper ends of the simulated band the VSWR is very high, which confirms poor matching far from resonance. However, near the main resonance around 5.7–5.8 GHz, the VSWR drops to a minimum value close to 1–2, indicating that the input impedance of the antenna is close to $50\ \Omega$. This behaviour is consistent with the S11 result in Figure 4.1 and confirms that the circular patch is well matched around its design frequency.

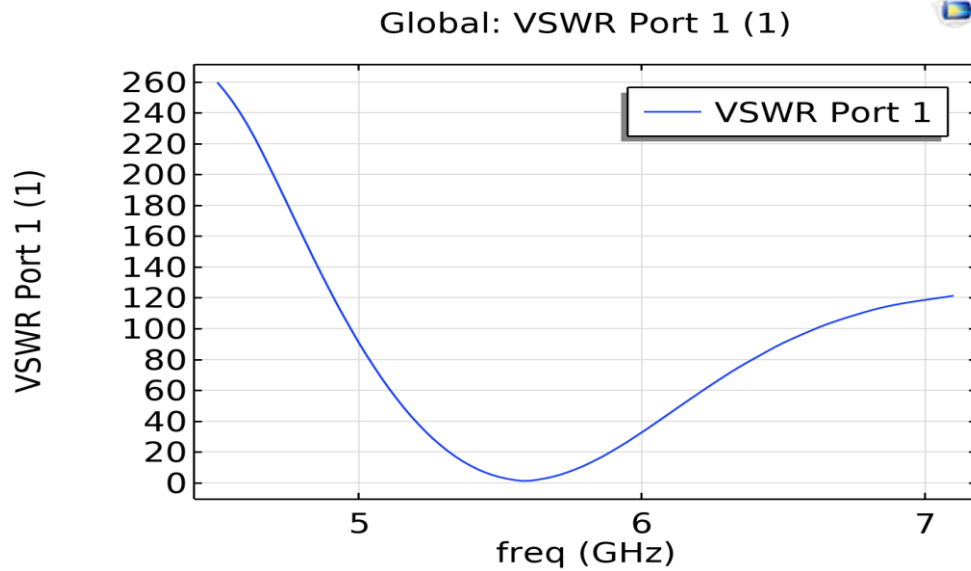


Figure 4.1.2

C. Radiation pattern

The three-dimensional far-field radiation pattern at the resonant frequency is presented in Figure 4.1.3. The antenna produces a single broad main lobe directed away from the patch surface, with lower field levels towards the back side. This is typical for a microstrip patch and shows that the antenna radiates mainly in the broadside direction, which is suitable for applications where coverage in front of the antenna is required.

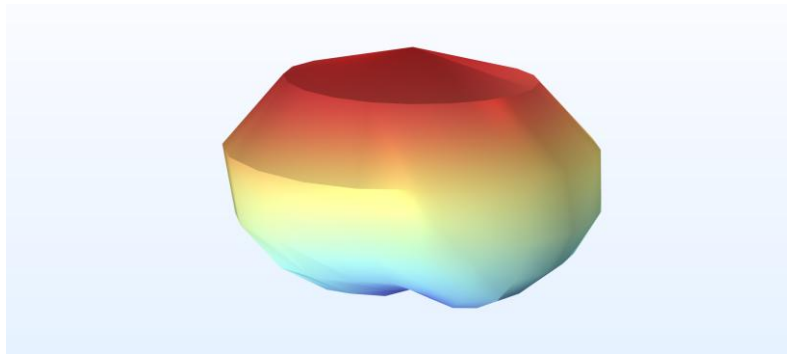


Figure 4.1.3

D. Gain

The maximum gain versus frequency for the circular patch is plotted in Figure 4.1.4. Across the simulated band, the gain gradually increases from about 4.8 dBi at the lower frequencies to a peak of approximately 6.9 dBi near 5.8–5.9 GHz and then decreases again as the frequency moves away from resonance. The gain value at the design frequency is therefore around 7 dBi, which is

reasonable for a single microstrip patch and confirms that the antenna can concentrate radiation in the broadside direction with moderate directivity.

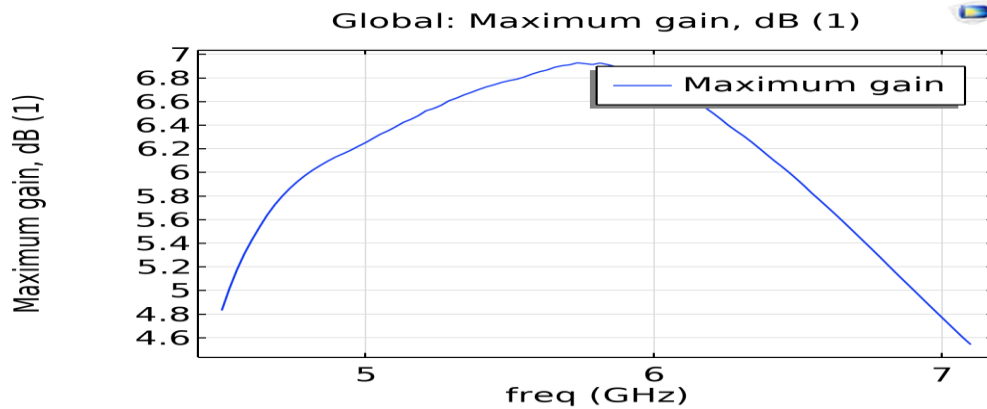


Figure 4.1.4

4.2 Dual-Port PIFA Antenna

. The antenna is printed on a finite ground plane with a raised top plate and two feeds on opposite sides, creating two input ports for diversity operation. The performance is discussed in terms of the reflection coefficient, impedance matching, radiation pattern, and gain.

A. Reflection coefficient

The reflection and transmission coefficients (S_{11} , S_{22} , and S_{21}) are plotted in Figure 4.2.1 over the 1–4 GHz band. Port 1 shows a clear resonance around 2.3–2.4 GHz, where S_{11} reaches a minimum of about -30 dB. Port 2 also has good matching in the same band, with S_{22} staying below roughly -10 dB across a wide frequency range. The transmission coefficient S_{21} , which represents coupling between the ports, remains much lower than the reflection terms (around -15 dB or below in most of the band). This indicates reasonable isolation and limits mutual coupling between the two ports.

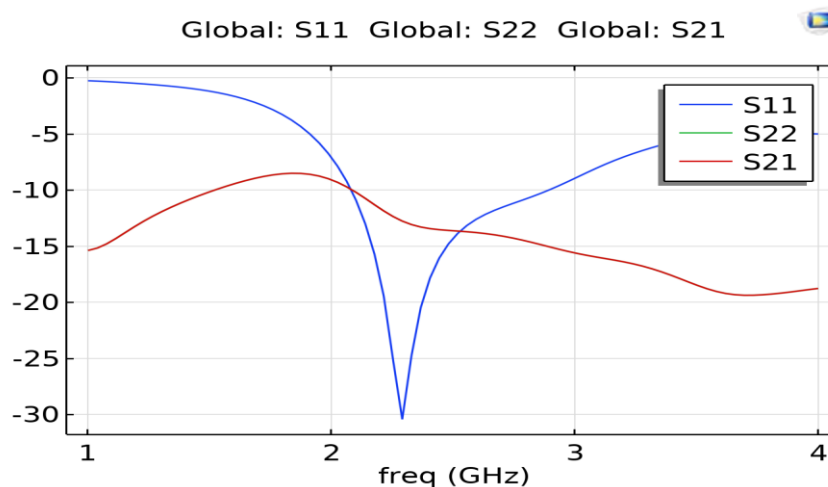


Figure 4.2.1

B. Impedance matching

Impedance matching can be seen more clearly from the VSWR curves in Figure 4.2.2 For Port 1, the VSWR starts very high at 1 GHz but drops quickly and reaches a value close to 1 near the resonant frequency around 2.3–2.4 GHz, showing that the input is well matched to a 50 Ω feed. The VSWR for Port 2 is also acceptable over the main operating band and stays close to or below 2. These results confirm that the designed PIFA is effectively matched at its working frequencies, so only a small fraction of the power is reflected along the feed line.

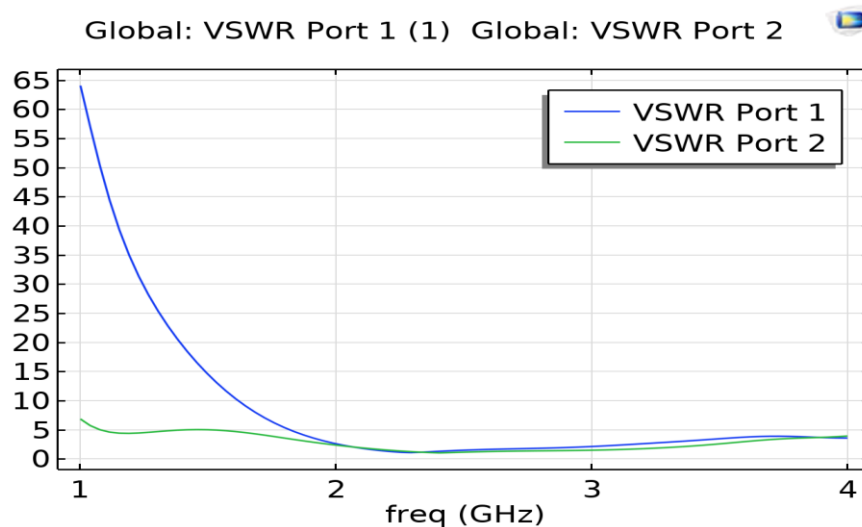


Figure 4.2.2

C. Radiation pattern

Figure 4.2.3 displays the three-dimensional far-field radiation pattern at 2.443 GHz. The antenna produces a single main lobe directed away from the ground plane, with relatively low side-lobe levels. The pattern shows that the antenna has good coverage around the broadside direction, which is suitable for portable and handheld devices that need reasonably wide angular coverage

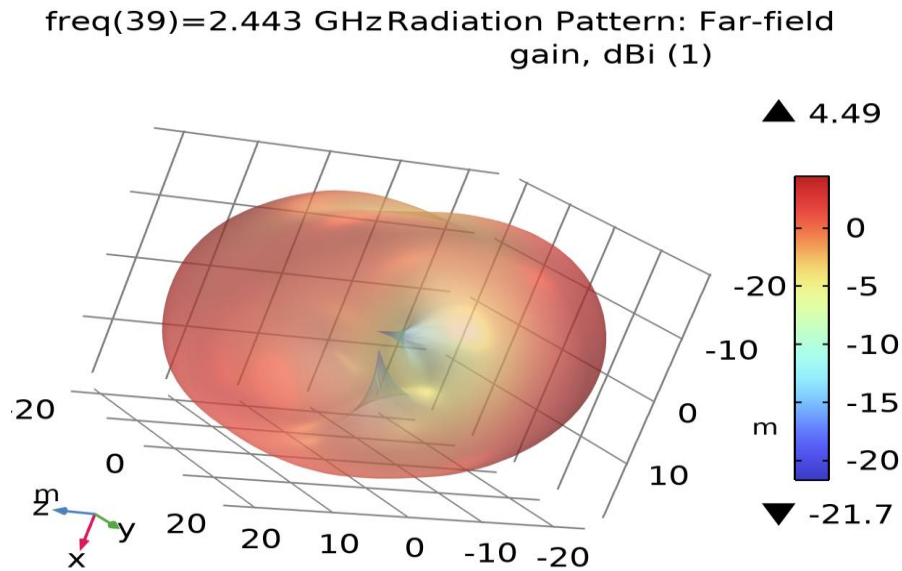


Figure 4.2.3

D. Gain

The simulated maximum gain versus frequency for both ports is plotted in Figure 4.2.4. Around the main operating band near 2.4 GHz, the gain is approximately 4–5 dBi, and the 3D radiation plot at 2.443 GHz reports a peak gain of about 4.49 dBi. As the frequency increases towards 4 GHz, the gain rises further and reaches values of about 7–8 dBi. This shows that the PIFA can concentrate radiation in the desired direction and provide sufficient gain for short-range wireless applications such as WLAN or Bluetooth-type systems.

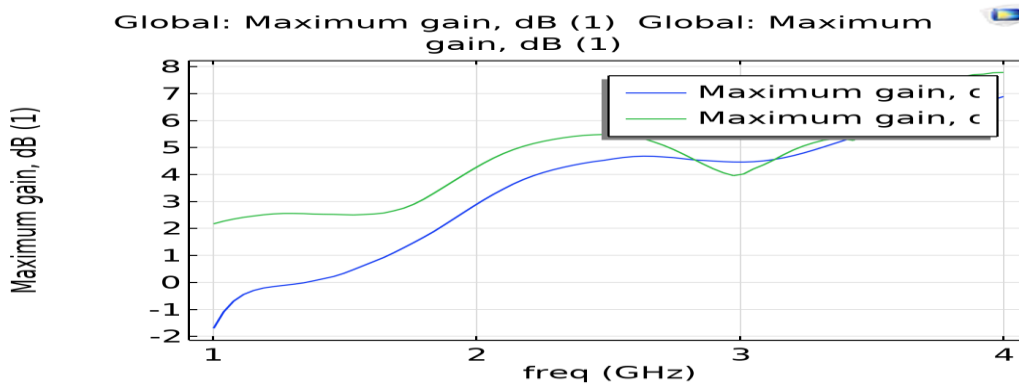


Figure 4.2.4

4.3 18 GHz Rectangular Microstrip Linear Array

A. Reflection coefficient

The simulated reflection coefficient S11 of the array is plotted in Figure 4.3.1. A clear resonance is observed in the upper part of the band, with a minimum S11 value of roughly -20 dB around

18–19 GHz. Over this region, the reflection coefficient stays below -10 dB, which indicates good impedance matching between the array input and the 50Ω feed line. Away from the resonance, S11 increases to 0 dB, meaning that more power is reflected and the array is not efficiently excited.

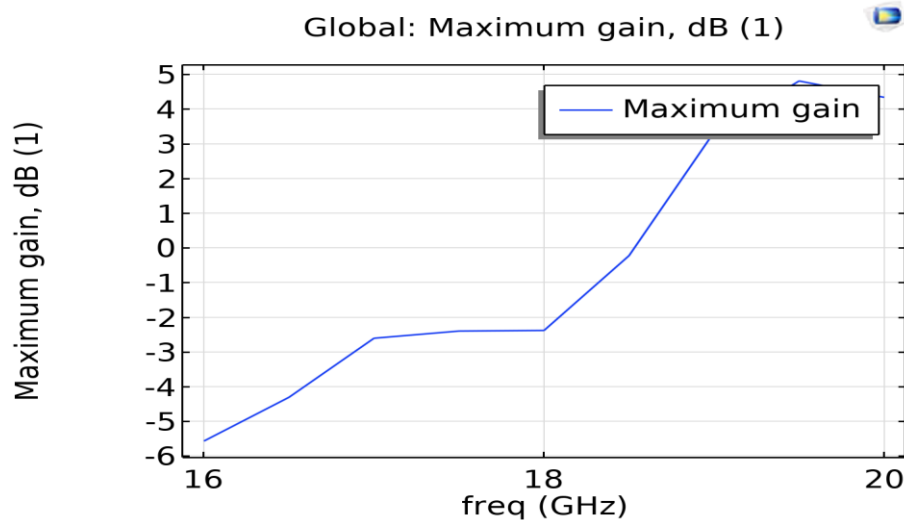


Figure 4.3.1

B. Impedance matching (VSWR)

The corresponding VSWR for Port 1 is shown in Figure 4.3.2. Between 16 GHz and 20 GHz the VSWR gradually decreases from values slightly above 2 at the lower end of the band to a minimum of about 1.2–1.3 near the resonance region. This minimum VSWR confirms that the array is well matched around 18–19 GHz, where the input impedance is close to 50Ω . The VSWR values remain below 2 over most of the operating band, which is acceptable for practical high-frequency applications.

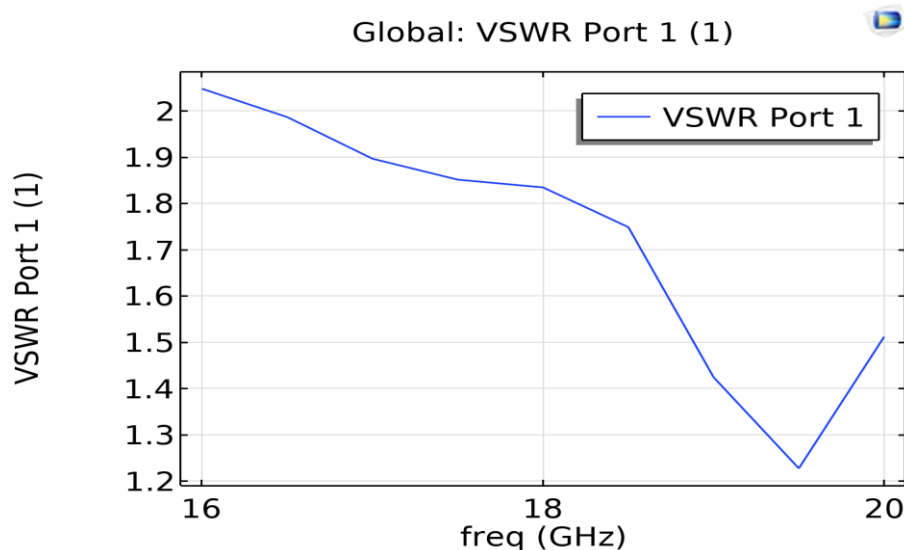


Figure 4.3.2

C. Radiation pattern

The three-dimensional far-field radiation pattern of the array at the design band is shown in Figure 4.3.3. The pattern exhibits a main lobe pointing approximately normal to the array surface, with a narrower beam compared to the single-patch antennas. The radiation is more concentrated in one direction, and the side-lobe levels are clearly lower than the main beam, which is characteristic of a linear array with improved directivity. This behaviour confirms that the 18 GHz array can provide directional coverage suitable for point-to-point or beamformed links.

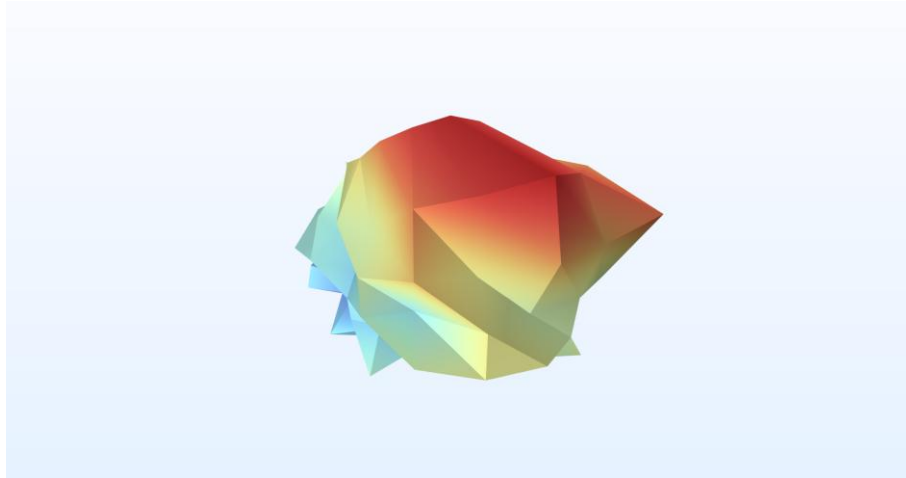


Figure 4.3.3.

D. Gain

The maximum gain versus frequency for the array is plotted in Figure 4.3.4. At 16 GHz the gain is relatively low (around -5 to -4 dB), but it increases steadily with frequency and becomes positive near 1 GHz, and then slightly decreases towards 20 GHz. This trend shows that the array provides higher gain around its matched band, where the radiation is more focused in the main lobe direction. Although the simulated gain is modest compared with large arrays, it is clearly higher than that of a single patch and demonstrates the benefit of using a linear array configuration

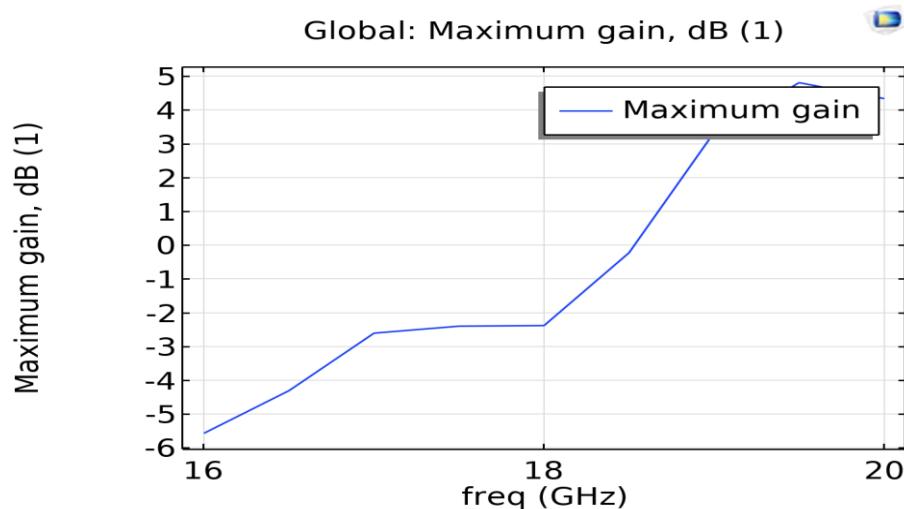


Figure 4.3.4

4.4. Random Forest Model Results

The Random Forest (RF) model performed well across most antenna parameters. The predicted-versus-actual comparison in Figure 4.4.1 shows that S_{11} , VSWR, input impedance, gain, TRP and efficiency closely follow the 45° line, with R^2 values mostly between 0.97 and 0.99. Directivity shows a wider spread of points, reflected in its lower R^2 (≈ 0.69), but the remaining metrics demonstrate strong agreement with the COMSOL data.

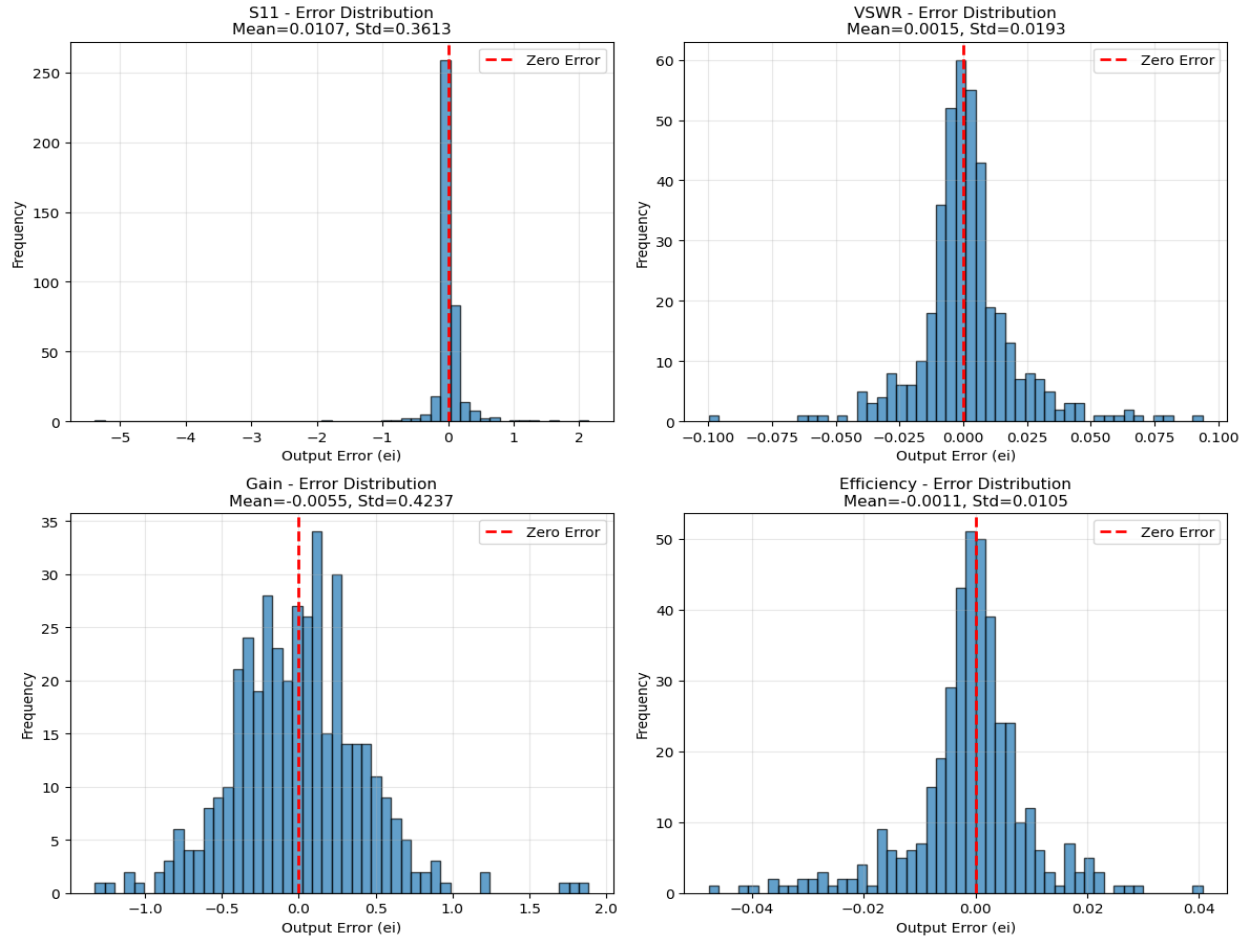


Figure 4.4.1

Figure 4.4.2's error-distribution plots confirm that the model errors centre near zero for S , VSWR, gain, and efficiency. The standard deviation is small for all outputs (≈ 0.36 dB for S_{11} , 0.02 for VSWR, 0.42 dB for gain and 0.01 for efficiency), meaning the model is stable and not biased in any direction.

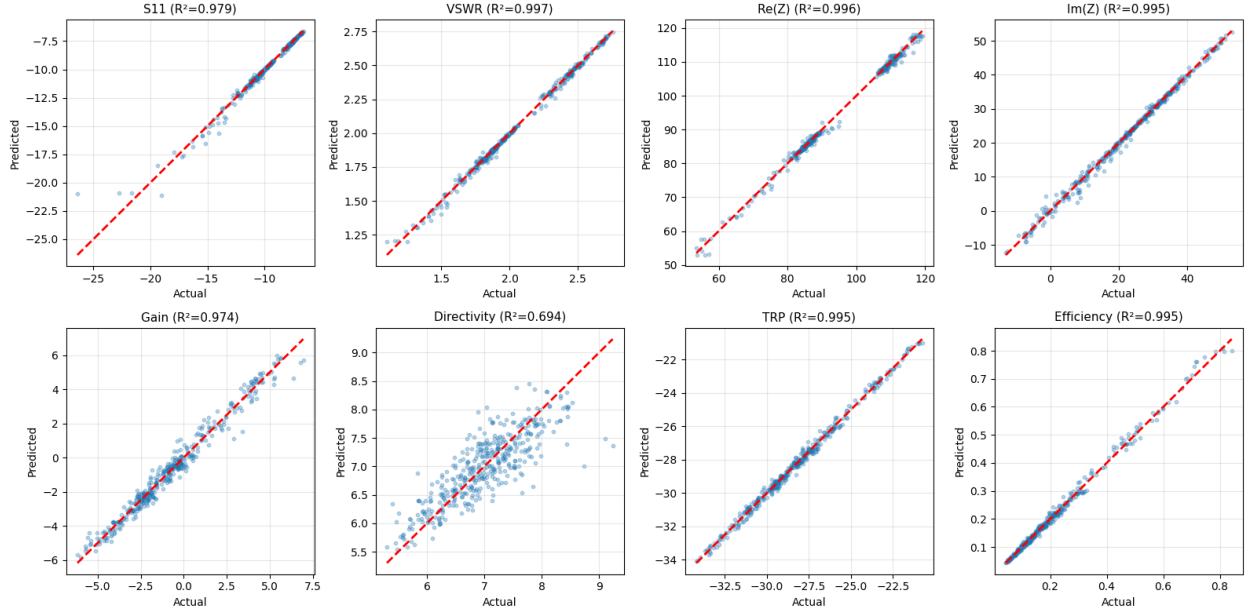


Figure 4.4.2

A single verification case was also checked using a design with $L_p = 3.9$ mm and $W_p = 5.89$ mm, operated at 18.878 GHz. The RF predictions showed an average error of 9.3%, with very accurate results for VSWR, TRP, and directivity. S_{11} and gain were predicted with acceptable accuracy, while $\text{Im}(Z)$ showed the largest deviation but remained within a usable range for fast optimisation.

Overall, the Random Forest model provides a reliable surrogate that predicts most antenna characteristics with adequate accuracy and is significantly faster than running a full COMSOL simulation.

4.5 Support Vector Regression (SVR)

The SVR was tested using the same data and geometry parameters as in the Random Forest model. These parameters included most of the antenna parameters predicted by the model, such as S_{11} , VSWR, real, and imaginary impedance. The predicted data from the SVR model has shown a linear correlation with COMSOL in the resulting regression plots.

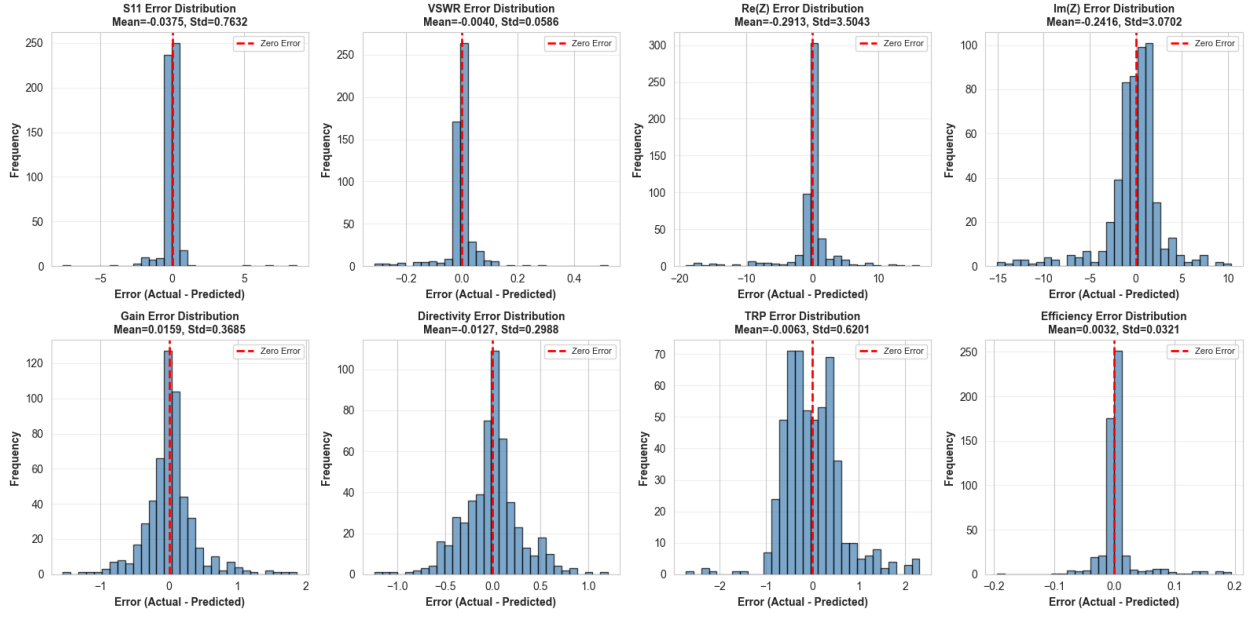


Figure 4.5.1 – SVR Regression Performance

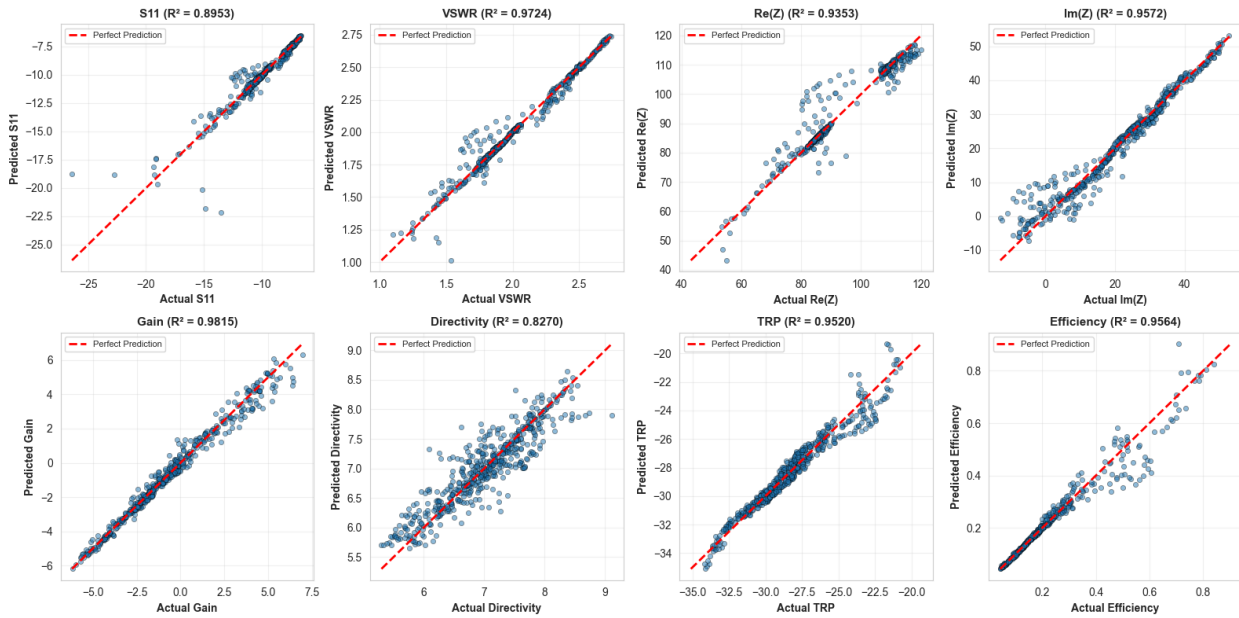


Figure 4.5.2 SVR predicted vs actual plots

The main issues in the SVR model were directivity and efficiency, where there was a greater spread of the prediction. But the S11 parameters, VSWR, and impedance characteristics agreed well, which denotes that the SVR model works satisfactorily in primary parameters

Key observations:

- S11 and VSWR strongly agree with COMSOL outputs.
- The real and imaginary impedance can be predicted correctly.
- Gain predictions are acceptable but slightly less stable.
- Directivity continues to be the weakest link in the SVR model.
- It is much slower than both ANN and RF when working with non-linear kernels.

In summary, the performance of SVR in impedance-related predictions is good but decreases in gain-related tasks.

4.6 Artificial Neural Network (ANN) Results

The ANN model presented the most robust performance out of the three ML models. The model, which was based on a hidden three-layer ANN, was able to effectively learn nonlinear associations between the geometry parameters and the antenna's behaviour.

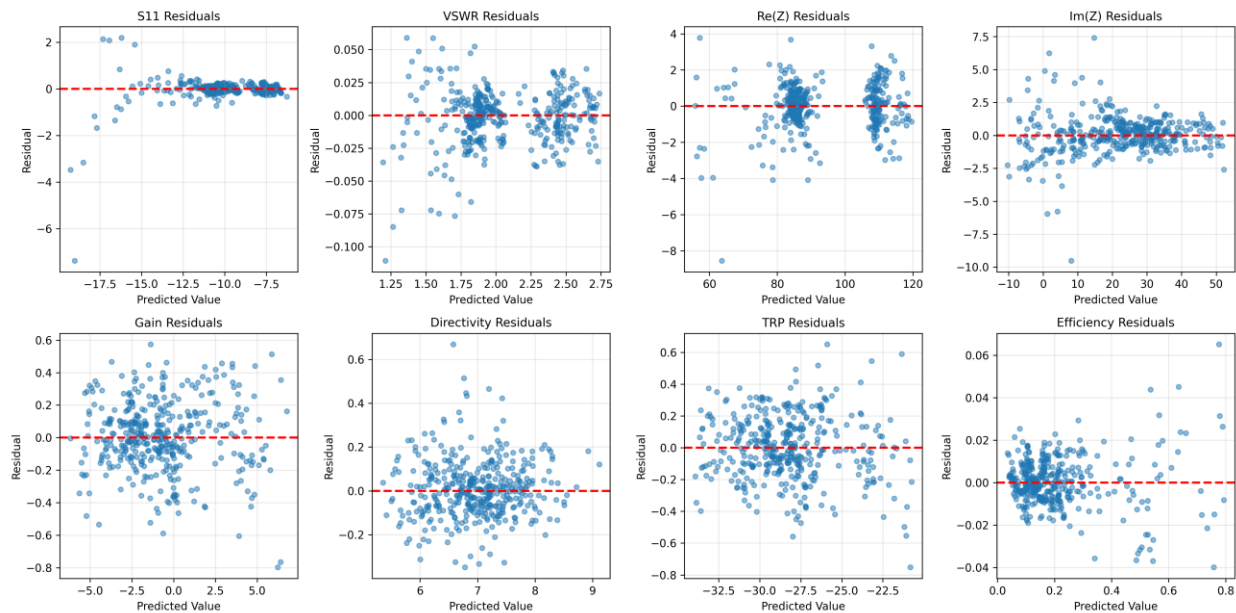


Figure 4.6. ANN regression residual plot

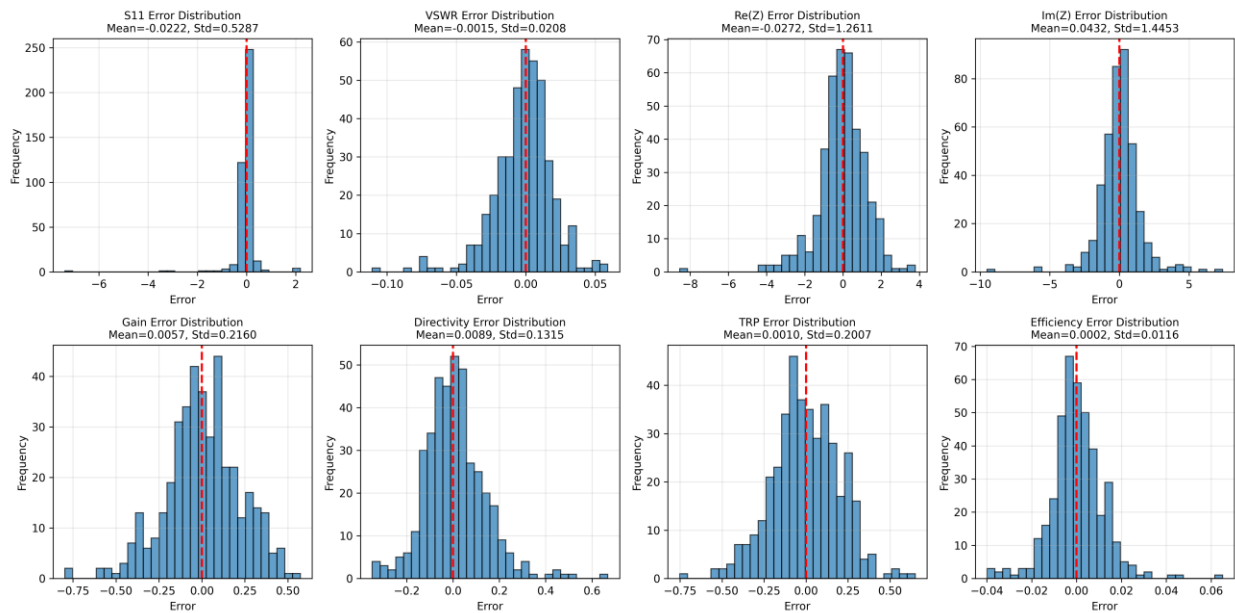


Figure 4.6.2 ANN error Distributions

It in fact produced:

- R^2 greater than 0.97 for most parameters (S11, VSWR, Re(Z), Im(Z), Gain)
- The lowest overall prediction error of all models:
- Stable predictions at high frequencies (17–20 GHz range)
- Best accuracy regarding S11 and VSWR, which play an important role in antenna matching
- Directivity had somewhat larger error ($R^2 \approx 0.69$), but was usable nevertheless

Key observations:

- ANN performed very effectively on non-linear data.
- The error values remained consistently centred at zero.
- Gain, S11, VSWR, TRP, and impedance predictions all had high accuracy.
- It trains quickly and performs well after that.
- ANN was the most computationally intensive model, although it was much faster at prediction times. The ANN model works best in predicting new antenna designs without undertaking COMSOL simulations.

4.7 Discussion

In conclusion, based on the above analysis, the performance of the ANN model appears to outpace all other candidates in terms of accuracy, with the second-best performer being the Random Forest model, while the worst performer was supporting vector regression (SVR). It can therefore be deduced that the ANN model stands out as the optimal model of choice in future optimisation, having in mind the plan to quickly search through the solution space and predict the performance characteristics of the antenna based on the new parameters. The second-best model in the analysis was the Random Forest model. The use of these models makes it possible to reduce the number of COMSOL simulations needed. Rather than having to simulate new designs, machine learning algorithms can first help in the screening of geometries such that only the designs identified as promising need to go through the process of analysis in COMSOL.

5. Conclusion

In this work, it has been shown that machine learning can act as an efficient surrogate model of the complete wave electromagnetic simulator in microstrip antenna array design and optimization tasks. To produce the data, a COMSOL Multiphysics model of an 18 GHz rectangular linear microstrip array was used to produce a dataset, which included geometries and corresponding performance parameters such as S11, gain, VSWR, and so on. These data were then used to train and test the performance of three different supervised algorithms—namely, Artificial Neural Networks, Support Vector Regression Model, and the Random Forest Regressor—on these tasks, where the ANN model resulted in the most accurate predictions with R^2 scores close to 0.98 in the test data, while the performance of the second and third algorithms was less accurate in comparison.

These results are in line with the existing literature, where ANNs and ensemble methods are shown to emerge as strong contenders in dealing with the complex nonlinear relationships from antenna geometry to radiation characteristics [9], [18], [19]. The trained model can offer predictions almost instantly after the dataset has been produced, unlike the execution of a complete series of new COMSOL simulations. In summary, the obtained output confirms that AI models can effectively model the dominant behaviour pattern of the 18 GHz microstrip array and hence can eliminate the need to perform an extensive number of full-wave COMSOL simulations during the design phase.

5.1 Future work

Extending the current work can take many forms. One would be to increase the size of the dataset and the number of antennas. In the current work, the aim was to verify the performance of the network using data related to an individual 18 GHz rectangular linear array of antennas, characterised by designs of single element

circular patch antennas and PIFA antennas. Later, it can cover many geometries, like patch geometries of other shapes, slot antennas, dipole-based arrays, and multiband antennas, and can shift from the current sub-6 GHz to millimetre wave frequencies. Datasets comprising many

geometries and multiband antennas have proved fruitful in many recent works related to the design of antennas using machine learning algorithms for patch antennas and MIMO antennas [18], [19].

One other area would be to investigate some of the more complex ML architectures and algorithms developed in 'reverse' design. In the current project, we have effectively applied these algorithms in a forward manner. More recently, it has been demonstrated that the reverse process can in fact be facilitated using deep neural networks, in which case the network proposes suitable antenna dimensions based directly upon target requirements such as resonant frequency or desired gain [20], [21]. Utilising the above process in its current form and including reverse design will enable the target performance requirements to produce suitable geometries, which can in turn be tested in COMSOL.

Finally, based on the concept already applied in the poster of the ongoing project, the long-term aim would be the creation of an interactive design assistant using AI. In this assistant, the user could define the high-level requirements (e.g., working frequency range, desired gain value, maximum size, and substrate constraints), and based on these, the assistant would generate antenna parameters and designs along with the S_{11} and gain plots. The ANN model developed in the course of the current work can serve as the backbone of such an assistant, and in the future, applying deep learning and reverse design can make such an assistant much more flexible, thereby shifting from basic academic research to something really practical to aid antenna designers in solving real-world design challenges.

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